

three-engined airliners competing for the same market niche; they inevitably ran into commercial difficulties since there was not enough room for both of them. Lockheed, which came off the worst in this contest, never made an airliner again. The Airbus A300, on the other hand, was a major success. With only two engines and two crew in the cockpit, it marked a significant step forward in economy of operation. By the 1980s Airbus Industrie had established itself as Boeing's most vigorous competitor on the world market.

Increased competition

The introduction of the wide-bodied jets was not an immediate commercial success. From 1973, economic recession slowed the rise in passenger numbers just as oil price hikes made the airliners vastly more expensive to fly – a 747 uses more than 3,000 gallons (13,500 liters) of fuel an hour. The new 747s and DC-10s often flew three-quarters empty, giving passengers the temporary advantage of impressively spacious accommodation. Some desperate airlines used the cavernous space to provide airborne piano bars or restaurants. Once the recession was over, of course, airlines went back to packing in passengers, sacrificing space and comfort to hold down prices.

The difficult years of the 1970s left some airlines with precarious finances – in 1975 TWA



ECONOMICAL AIRBUS

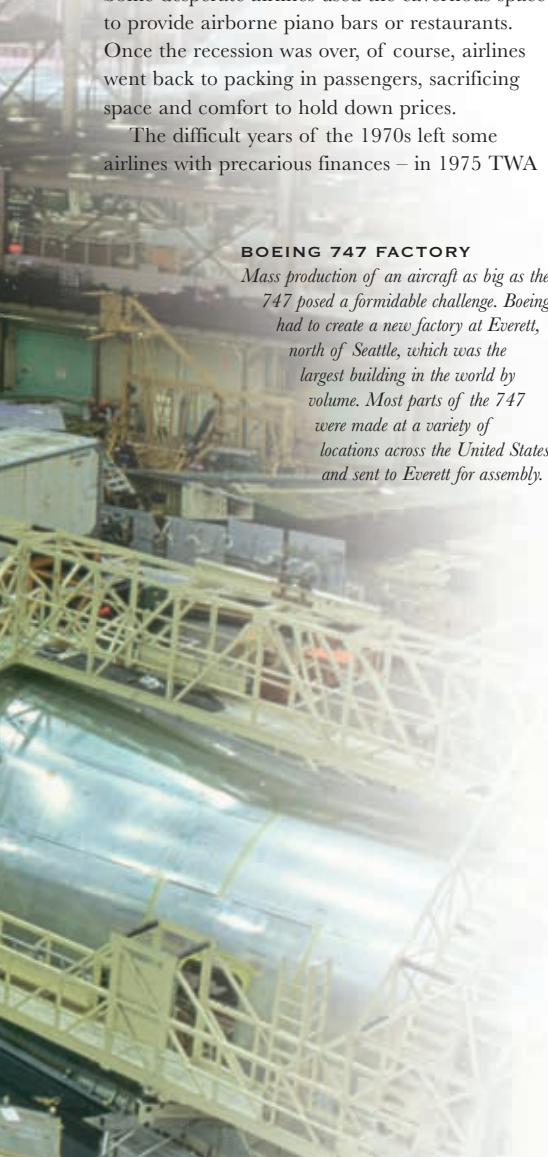
Airbus Industrie shook the airliner market with the introduction of the A300, beginning service in 1974. It was the first widebody airliner to fly with only two engines and two cockpit crew. This made all its variants impressively economical to operate.

had to sell some of its 747s to Iran to raise the cash to pay its staff's wages. Another blow to the long-established major carriers was the progress of deregulation from the late 1970s. This was primarily an American phenomenon, although mirrored to a degree worldwide. Free (or at least freer) competition on air routes forced prices down further and soon brought some famous casualties, notably Braniff in the 1980s, and then PanAm, which stopped services in 1991. Even in Europe, where national support for flagship airlines continued, the irrepressible rise of charter flights undercut the official carriers. Small operators were able to buy perfectly viable second-hand airliners, as the major airlines felt forced to renew their fleets long before their existing aircraft had exhausted their usefulness. The airline business had also long ceased to be largely the preserve of Europe and North America, with companies such as Singapore Airlines and Cathay Pacific becoming among the world's most prestigious airlines.

From the 1970s jet flight became an experience open to everyone. As such it inevitably lost its connotations of glamour, romance, or excitement. The European consortium's choice of the name "Airbus" for its product spoke volumes about the prosaic nature of air travel in the age of the wide-bodied jets. Most passengers on a 747 did not even have a window. Much effort was devoted to distracting passengers from the fact that they were flying at all, insulating them with in-flight entertainment and reducing them to passive consumers of duty-free goods, snacks, and drinks. One jaundiced journalist referred to modern air travel as "the most constrained form of mass transport since the slave ships."

BOEING 747 FACTORY

Mass production of an aircraft as big as the 747 posed a formidable challenge. Boeing had to create a new factory at Everett, north of Seattle, which was the largest building in the world by volume. Most parts of the 747 were made at a variety of locations across the United States and sent to Everett for assembly.



DC-10 ENGINE

The fan-jet engines used to power widebody jetliners are astonishing pieces of equipment. A turbofan such as this one from a DC-10 can generate over 50,000lb (25,000kg) of thrust – almost three times the thrust of each engine on a 1960s Boeing 707. The extra power translates into increased aircraft size and payload.

